Geospatial Deep Learning for Road Network Detection: A Literature Review

Kelvin Wei
University of Cape Town
Cape Town, South Africa
wxxkel001@myuct.ac.za

Abstract

The extraction of accurate and up-to-date road network data is essential for downstream geospatial analyses, including infrastructure planning and environmental monitoring. While deep learning has achieved significant success in road extraction, current methodologies predominantly rely on costly, high-resolution commercial remote sensing imagery. This reliance creates a distinct research gap, systematically underrepresenting rural road networks in developing nations. This literature review evaluates the viability of using freely available, medium-resolution satellite imagery (10 m – 30 m), such as Sentinel-2 and Landsat data, for road network detection. We examine the inherent challenges of lower spatial resolution, notably the mixed pixel problem and environmental occlusions, alongside the specialised frameworks required to process geospatial metadata. Furthermore, this review assesses current deep learning methodologies, ranging from supervised segmentation models (e.g., U-Net, D-LinkNet) and super-resolution techniques to the recent emergence of large geofoundational models (e.g., Road-SAM, TerraMind). Ultimately, we highlight that the fusion of multi-spectral bands, coupled with multi-modal foundational models, presents a highly promising avenue for overcoming resolution limitations and democratising global road network mapping.

Keywords

Satellite Imagery, Computer Vision, Road Detection

1 Introduction

Geospatial data is information related to a geographical location on the Earth's surface. Road networks and satellite imagery are examples of geospatial data that is commonly visualised on maps. The procurement of up-to-date and accurate road networks data enables various downstream analyses like network density estimation which assists in environmental monitoring, infrastructure planning and route optimisation. Deep learning methods has been a successful way to extract road networks and has seen significant advancements and results. However, road extraction with deep learning is frequently accomplished with high resolution remote sensing imagery [23]. The acquisition of large volumes of high resolution data is limited by cost and is heavily commercialised. Satellite data is a promising alternative source of open licensed data that has become available in large volumes due to various space agency earth observation programs such as the Copernicus [10] and Landsat [24] programs. Satellite images provide global coverage on valuable temporal insights, due to their frequency satellite revisit times around the globe, and include multispectral image bands. These benefits are valuable for near real-time and historical road network monitoring in under represented rural areas in development countries.

1.1 Inherent Challenges

Road detection using open licensed satellite data is a challenging task, in comparison to using high resolution remote sensing data, due to their relatively low spatial resolution. A mixed pixel problem is when a single pixel covers multiple feature edges. Low resolution hinders a model's ability to classify fine-grained features [13] due to the mixed pixel problem. Edges of fine-grained features like narrow roads are commonly not visually detectable by a human [11] due to a single pixel capturing the road and its nearby features.

Rural road detection in developing countries are under-represented and presents unique challenges. Road detection models commonly focus on urban road detection and do not generalise well to rural settings. Rural road detection is challenging as they are typically narrower than urban roads, unpaved and have high curvature variations. Models must account for the diverse material compositions and varying properties inherent in unpaved road surfaces. The problem is exacerbated by the lack of quality rural road datasets [22].

Cloud cover and shadows are a common problem present in remote sensing data. Sensors on the satellite are be blocked by clouds which reveals partial/no information about the land it's capturing. Data with cloud cover typically require a preprocess filtering step with Fmasking being the most common method [16]. Filtering introduces a potential mismatch in temporal coherent of dynamic features when multiple satellite images are merged to create a cloud free image [24].

1.2 Aim

The remainder of the review is structured as follows. Section 2 discusses various common methods that utilities the properties of satellite data to assist in detecting road networks. Section 3 introduces deep learning methods that are used in detecting road networks. Section 4 highlights promising work in the detection of road networks from medium resolution satellite imagery.

2 Geospatial Data

2.1 Frameworks

Frameworks allow deep learning methods to utilise the properties of geospatial satellite data. The application of deep learning methods to geospatial satellite data is a non-trivial task, as existing deep learning libraries are not designed to handle the unique properties of geospatial data [19]. Geospatial data contains metadata about it's geographical location, time and multispectral bands which may

be distributed over many different file formats. Satellite images are a 2D projection of earth's 3D surface using a coordinate reference system (CRS). Datasets may use different CRSs which need to converted into a common CRS for deep learning manipulation using specialised libraries (GDAL).

Models trained on geospatial data do not publish their data pipelines [25]. Transparent data pipelines are important to ensure reproducibility of models and techniques. Deep learning geospatial frameworks are not common, but they are essential in assisting with common complex tasks. The creation of road datasets for deep learning tasks requires pairing cloud free satellite imagery and road data. Road datasets are commonly in vector formats. Raster mask labels need to be generated by projecting road vector coordinates to mask out corresponding pixels and additional processes to ensure high quality data.

InstaGeo [25] is a recent open source end-to-end framework which handles data acquisition and preprocessing, model training and inference and data visualisation. Instageo's data model can be utilised for inference or recreating datasets. InstaGeo utilises a SpatioTemporal Asset Catalog (STAC) API which allows the framework to query, filter and retrieve satellite granules. InstaGeo's model component can distillation geofoundational models in lightweight task-specific models that retain similar performance to the original model using a teacher student architecture.

2.2 Multi-Spectral Data

Satellites can carry various sensing instruments that capture different aspects of the Earth's surface. The sensors relevant to road detection can be broadly classified into optical, multispectral, and radar sensors. Sentinel-2 and Landsat 8/9 satellites have optical sensors that capture light in the visible and near infrared range at a spatial resolution of 10 m and 30 m per pixel, respectively [9]. Combinations of bands have been found to be useful in the detection of roads. Gašparović and Tomislav [6] demonstrate that supervised land cover classification benefits from image fusion methods utilising certain combinations of bands. They conclude that fusion methods (P + XS and PCA) improve the classification of vegetation, bare soil and coastlines. Fusion methods are promising, as roads in rural areas are frequently surrounded by vegetation.

2.3 Super Resolution

Freely available satellite data suffers from a mixed pixel problem due to an insufficient resolution to detect fine grained road networks. Li et al. [13] concluded that for the task of land classification, the optimal resolution is 4-6 m, based on classification accuracy. The authors argue that the models pick up complex fine-grained details, which increases intra-class variance and decreases classification accuracy. Whereas, low-resolution images (10/30m) introduces a mixed pixel problem.

Super resolution utilities deep learning to enhance low resolution images. Single image super resolution (SISR) does this by using independent images and multiple image super resolution (MISR) utilises relationship between multiple images to enhance images [20]. SISR was used by Jia et al. [8] to improve road network detection by super resolving Sentinel 2 imagery (10 m) to 2 m. Their technique incorporates relative positions of positions between roads and rural settlements with SISR to achieve a road detection.

Freely available Sentinel 2 satellite data have a 5 day revisit interval with 10m per pixel spatial resolution for certain bands [9]. The high revisit rate can be exploited in MISR to improve road network detection. Sirko et al. [18] used MISR with 32 multiple overlapping Sentinel 2 images to train a model to detect road networks and buildings. A teacher-student architecture was used. The teacher model which was trained on high-resolution images to train a student model, which sees multiple equivalent low-resolution images to detect high resolution features.

Super resolution is prone to undesirable artefacts like hallucinating non-existent structures [1]. As a result, super resolution methods are paired with additional grounding information to successfully improve road network detection.

3 Deep Learning Methods

3.1 Evaluation

The task of road extraction can be broadly classified into road segmentation and road graph extraction [15]. Road extraction is fundamentally a pixel classification problem and evaluation is based on pixel-level metrics. Precision, recall, F1 score, and intersection-over-union (IoU) are the primary metrics used for pixel-level metrics [15]. Road segmentation tasks avoids accuracy as a metric as road data labels are heavily imbalanced and leads to inflated values.

Road graph extraction aims to extract the nodes and edges that define a road network and require graph-based vectorisation metrics for evaluation. Average path length similarity (APLS) and TOPO are commonly used evaluation metrics [15]. APLS [**?**] is a metric that measures the difference in optimal path lengths between nodes in the ground-truth graph and the predicted graph. TOPO [15] is a metric measures geometric accuracy and has a strong penalty for incorrect disconnections.

Common supervised loss function for road extraction are binary cross-entropy (BCE) and dice loss. Binary cross entropy (1) is done for each pixel in the predicted image. Dice loss (2) is noticeably effective in addressing road label imbalances as it measures the overlap between the predicted pixels and ground truth pixels.

$$L_{BCE}(P, Y) = -\sum_{i=1}^{W}\sum_{j=1}^{H}[y_{ij} \cdot \log p_{ij} + (1 - y_{ij}) \cdot \log(1 - p_{ij})] \quad (1)$$

$$L_{DCL}(P, Y) = 1 - \frac{2 \cdot |P \cap Y|}{|P| + |Y|} \quad (2)$$

In dice loss, P represents the predicted results and Y represents the ground truth. In BCE, W and H represents the width and height of the image.

3.2 Supervised Road Segmentation

CNN models are a popular architecture for detecting roads. CNN look at pixel-by-pixel comparisons and can create road segmentation masks with corresponding attributes of a road. Roads themselves have different attributes like width, surface type and direction

of travel. CNNs are known to have trouble with preserving the connectivity of roads [14]. U-Net [17] is a common baseline model used to detect medical and remote image segmentation tasks.

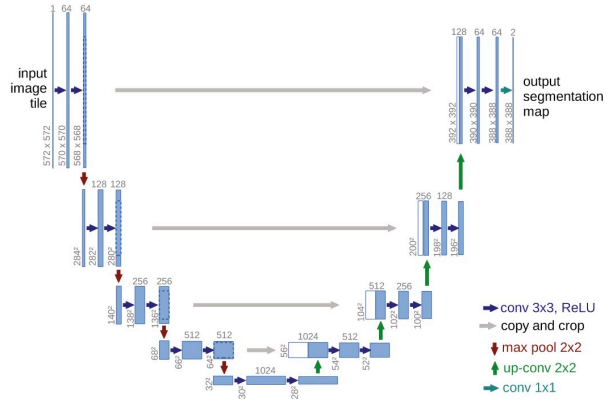


Figure 1: Unet Architecture [17]

D-Linknet [26] won the DeepGlobe Challenge [4]. The DeepGlobe dataset is a road dataset of containing high resolution images in rural and urban settings. D-Linknet makes modifications upon the Unet architecture for detecting long, thin, and continuous features. Dilated convolutions inserts gaps between kernel elements of a standard convolutional operation. D-Linknet addresses the issue of a limited receptive field by introducing dilated convolutions to increase the model's receptive field to capture long-range spatial dependencies.

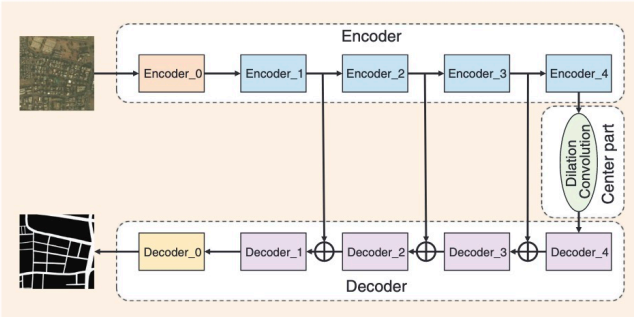


Figure 2: DLink structural framework [23]

3.3 Road Graph Extraction

Road topological maps are structured as directed graphs $G(V, E)$, where V represents the set of nodes (representing road intersections or endpoints) and E represents the edges (the road segments connecting them). Graph-based methods and models are naturally more suited to address the challenge of preserving topological connectivity. Road graph extraction techniques are categorised into segmentation-based methods and graph generation methods. Segmentation based methods initially generates segment maps which is later iteratively vectorised. Graph-based approaches directly generates road graphs rom satellite imagery. RoadTracer [2] is a segmentation-based method which utilities an iterative search algorithm that walks to generate a graph of the road network.

3.4 Foundational Models

The large volumes of freely available satellite data allows foundational models to be pre-trained on large volumes of unlabelled data with self-supervision. The model learns high-level representations of the structure and features of training data and generalises to diverse set of tasks. Generalisation to various tasks are done by further fine-tuning the model on task specific labelled data, which adapts the models to a desired task.

Prithvi-EO-2.0 [21] is a geospatial foundation model (GFM) pretrained on a massive global dataset of 4.2 million time-series samples from NASA's Harmonised Landsat-Sentinel-2 (HLS) archive at 30 m resolution. The model incorporates positional embeddings and data metadata. Prithvi is potentially not suitable for road extraction as it is pretrained on 30 m resolutoin images.

Success has been found in adapting the foundational model SAM [5] to road extraction. Road-SAM [5] is an adaptation of SAM designed for road extraction from very high resolution (VHR) optical images. A key innovation is the task-specific input module of Explicit Visual Prompting (EVP), which learns prompts from image embeddings to capture intricate road edges and textures. Road-SAM achieves state of the art performance while updating only 10% of the model's parameters. RN-SAM (Road Network-Aided SAM) [12] leverages OpenStreetMap (OSM) as a source of road network information to enhance SAM's performance. The framework operates in two phases: a pre-training phase where SAM is fine-tuned with road network-based point prompts, and a test phase that applies test-time optimisation using OSM data. During inference, the model minimises residuals between its predicted masks and the OSM road network, allowing for per-image refinement that significantly improves robustness in cross-region scenarios and under-occlusion conditions.

TerraMind is a recent foundational model that allows for multimodality input and output [7]. It is trained on TerraMesh dataset [3] which is currently the largest publicly available globally diverse multimodal EO dataset (14TB compressed). It contains 8 modalities, which include data from Sentinel 1 and 2 satellites (Optical + SAR), NDVI, Copernicus DEM and land-use/land-cover labels from ERSI. The dataset is designed for pre-training large foundational models like TerraMind [7].Satellite imagery can be presented in multiple modalities. Broadly, they can be classified as pixel-level, token-level and sequence input. Rasters of satellite imagery from Sentinel 1 and 2 constitute pixel-level input. Images to be tokenised to token-level inputs, which allows Terramind to use land-use/land-cover (LULC) maps, vegetation index (NDVI) and Digital Elevation Model (DEM) to learn high-level representations. Sequence input, such as image text caption and coordinate data, is information that is commonly associated with satellite imagery. TerraMind can consume and output different modalities of geospatial data. The model learns cross-modal relationships via correlation learning [7]. TerraMind's multi-modality capabilities allow for "Thinking-in-Modalities", which is a recursive inference technique that allows the

model to generate synthetic modalities to be then used in conjunction with the original input. Road detection tasks can potentially benefit from the generation of synthetic LULC maps.

4 Unsupervised Learning

Satellite imagery is typically abundant and unlabelled. Semi-supervised learning (SSL), weakly-supervised learning (WSL) and unsupervised learning (USL) are techniques that reduce the need for labelled data [15]. SSL utilises a small set of labelled samples with a large volume of unlabelled data. WSL reduces labelling efforts by utilising point based annotations or scribbles [**?**].

Such methods suffer from a performance degradation has been observed when the models is presented with data outside of it's training data [15].

5 Discussion

Much of current road network research is preformed on high-resolution remote sensing data (0.3m–1m) which is a resource which underrepresented rural communities in developing nations lack. There is a gap in the research in detecting roads from medium to low resolution data (10 m - 30) found in freely available satellite data from Sentinel 2 and Landsat satellites. Medium resolution road detection data is challenging task, namely due to the mixed pixel problem, that is worth investigating. There are benefits to utilising sentinel 2 data for road detection. Currently models are trained on road datasets that are biased toward urban and suburban environments in developed countries. In contrast, rural road networks, particularly in Africa, are largely ignored. Sentinel 2 satellite data provides global coverage and allow for near real time and historical monitoring of road networks. This allows for downstream analyses of how road networks densities affect nearby land use.

Geospatial foundational models on road network detection are promising. TerraMind is trained on Sentinel 1 and 2 data. It is also a multi-model foundation model which can be potentially beneficial in detecting road networks. Road-SAM shows promising techniques results and it a graph based approach is is promising preserving connectivity of road networks.

Various methods that utilities properties of satellite data is not commonly explored. Normally RGB of high resolution data is used in the detection of road networks. However, combinations of various bands is found to be useful in assisting in the detection of roads. Super resolution methods are also very promising provided that additional grounding information is provided to prevent hallucination

6 Conclusion

Freely available medium-resolution data, provided by Sentinel 2, offers a highly scalable alternative to expensive high-resolution data. However, leveraging this data requires overcoming significant hurdles.

Deep learning has demonstrably advanced the field of geospatial road extraction. However, the current literature is heavily biased toward high-resolution, urban datasets. This review has highlighted a gap in the utilisation of freely available, medium-resolution satellite imagery to address the significant under-representation of rural road networks, particularly in developing nations. Extracting fine-grained topological features from 10 m to 30 m resolution data introduces complex challenges, most notably the mixed pixel problem and the nuanced material compositions of unpaved roads.

Despite these obstacles, the integration of super-resolution techniques (SISR and MISR) and the strategic fusion of multi-spectral bands offer methods to enhance spatial fidelity and classification accuracy. Furthermore, new development of geofoundational models, such as TerraMind and Road-SAM, signals promising potential to road detection due to their multi-modal inputs, graph-based topological extraction and cross-modal correlation learning properties. Solving obstables in utiliting lower resolution satelliate imagery can enable near real-time, global monitoring of road infrastructure, ultimately facilitating critical downstream analyses for sustainable development and land-use planning.

References

[1] Cesar Aybar, Julio Contreras, Simon Donike, Enrique Portalés-Julià, Gonzalo Mateo-García, and Luis Gómez-Chova. 2026. A Radiometrically and Spatially Consistent Super-Resolution Framework for Sentinel-2. 334 (2026), 115222. doi:10.1016/j.rse.2025.115222

[2] Favyen Bastani, Songtao He, Sofiane Abbar, Mohammad Alizadeh, Hari Balakrishnan, Sanjay Chawla, Sam Madden, and David DeWitt. 2018. RoadTracer: Automatic Extraction of Road Networks From Aerial Images. 4720–4728. https://openaccess.thecvf.com/content_cvpr_2018/html/Bastani_RoadTracer_Automatic_Extraction_CVPR_2018_paper.html

[3] Benedikt Blumenstiel, Paolo Fraccaro, Valerio Marsocci, Johannes Jakubik, Stefano Maurogiovanni, Mikolaj Czerkawski, Rocco Sedona, Gabriele Cavallaro, Thomas Brunschwiler, Juan Bernabe Moreno, and Nicolas Longepe. 2025. TerraMesh: A Planetary Mosaic of Multimodal Earth Observation Data. 2419–2427. https://openaccess.thecvf.com/content/CVPR2025W/EarthVision/html/Blumenstiel_TerraMesh_A_Planetary_Mosaic_of_Multimodal_Earth_Observation_Data_CVPRW_2025_paper.html

[4] Ilke Demir, Krzysztof Koperski, David Lindenbaum, Guan Pang, Jing Huang, Saikat Basu, Forest Hughes, Devis Tuia, and Ramesh Raskar. 2018. *DeepGlobe 2018: A Challenge to Parse the Earth through Satellite Images.* arXiv.org. doi:10.1109/CVPRW.2018.00031

[5] Wenqing Feng, Fangli Guan, Chenhao Sun, and Wei Xu. 2024. Road-SAM: Adapting the Segment Anything Model to Road Extraction From Large Very-High-Resolution Optical Remote Sensing Images. 21 (2024), 1–5. doi:10.1109/LGRS.2024.3430900

[6] Mateo Gašparović and Tomislav Jogun. 2018. The Effect of Fusing Sentinel-2 Bands on Land-Cover Classification. 39, 3 (2018), 822–841. doi:10.1080/01431161.2017.1392640

[7] Johannes Jakubik, Felix Yang, Benedikt Blumenstiel, Erik Scheurer, Rocco Sedona, Stefano Maurogiovanni, Jente Bosmans, Nikolaos Dionelis, Valerio Marsocci, Niklas Kopp, Rahul Ramachandran, Paolo Fraccaro, Thomas Brunschwiler, Gabriele Cavallaro, Juan Bernabe-Moreno, and Nicolas Longépé. 2025. *TerraMind: Large-Scale Generative Multimodality for Earth Observation.* arXiv:2504.11171 [cs] doi:10.48550/arXiv.2504.11171

[8] Yuanxin Jia, Xining Zhang, Ru Xiang, and Yong Ge. 2023. Super-Resolution Rural Road Extraction from Sentinel-2 Imagery Using a Spatial Relationship-Informed Network. 15, 17 (2023), 4193. doi:10.3390/rs15174193

[9] Junchang Ju, Qiang Zhou, Brian Freitag, David P. Roy, Hankui K. Zhang, Madhu Sridhar, John Mandel, Saeed Arab, Gail Schmidt, Christopher J. Crawford, Ferran Gascon, Peter A. Strobl, Jeffrey G. Masek, and Christopher S. R. Neigh. 2025. The Harmonized Landsat and Sentinel-2 Version 2.0 Surface Reflectance Dataset. 324 (2025), 114723. doi:10.1016/j.rse.2025.114723

[10] S. Jutz and M. P. Milagro-Pérez. 2020. Copernicus: The European Earth Observation Programme. 56 (2020), V–XI. doi:10.4995/raet.2020.14346

[11] Avishkar Karnik, Shilpa Pimpalkar, Manjusree Panchagnula, and Subrat K. Acharya. 2025. Sentinel-2based Dataset Covering Indian Regions for Road Detection: A Dataset for Infrastructure Monitoring. 63 (2025), 112141. doi:10.1016/j.dib.2025.112141

[12] Ryosuke Kawamura, Pablo Guarda, Pradeep Narwade, Yash Patel, and Koichiro Niinuma. 2025. RN-Sam: Road Network-Aided Sam Optimization for Road Segmentation In Satellite Imagery. In *2025 IEEE International Conference on Image Processing (ICIP)* (2025-09). 1318–1323. doi:10.1109/ICIP55913.2025.11084282

[13] Runxiang Li, Xiaohong Gao, Feifei Shi, and Hao Zhang. 2023. Scale Effect of Land Cover Classification from Multi-Resolution Satellite Remote Sensing Data. 23, 13 (2023), 6136. doi:10.3390/s23136136

[14] Ruyi Liu, Junhong Wu, Wenyi Lu, Qiguang Miao, Huan Zhang, Xiangzeng Liu, Zixiang Lu, and Long Li. 2024. A Review of Deep Learning-Based Methods for Road Extraction from High-Resolution Remote Sensing Images. 16, 12 (2024). doi:10.3390/rs16122056

[15] Xiaoyan Lu and Qihao Weng. 2025. Deep Learning-Based Road Extraction from Remote Sensing Imagery: Progress, Problems, and Perspectives. 228 (2025), 122–140. doi:10.1016/j.isprsjprs.2025.07.013

[16] Darius Phiri, Matamyo Simwanda, Serajis Salekin, Vincent R. Nyirenda, Yuji Murayama, and Manjula Ranagalage. 2020. Sentinel-2 Data for Land Cover/Use Mapping: A Review. 12, 14 (2020), 2291. doi:10.3390/rs12142291

[17] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (Cham, 2015), Nassir Navab, Joachim Hornegger, William M. Wells, and Alejandro F. Frangi (Eds.). Springer International Publishing, 234–241. doi:10.1007/978-3-319-24574-4_28

[18] Wojciech Sirko, Emmanuel Asiedu Brempong, Juliana T. C. Marcos, Abigail Annkah, Abel Korme, Mohammed Alewi Hassen, Krishna Sapkota, Tomer Shekel, Abdoulaye Diack, Sella Nevo, Jason Hickey, and John Quinn. 2024. *High-Resolution Building and Road Detection from Sentinel-2.* arXiv:2310.11622 [cs] doi:10.48550/arXiv.2310.11622

[19] Adam J. Stewart, Caleb Robinson, Isaac A. Corley, Anthony Ortiz, Juan M. Lavista Ferres, and Arindam Banerjee. 2022. *TorchGeo: Deep Learning With Geospatial Data.* arXiv:2111.08872 [cs] doi:10.48550/arXiv.2111.08872

[20] Hu Su, Ying Li, Yifan Xu, Xiang Fu, and Song Liu. 2025. A Review of Deep-Learning-Based Super-Resolution: From Methods to Applications. 157 (2025), 110935. doi:10.1016/j.patcog.2024.110935

[21] Daniela Szwarcman, Sujit Roy, Paolo Fraccaro, Þorsteinn Elí Gíslason, Benedikt Blumenstiel, Rinki Ghosal, prefix=de useprefix=true family=Oliveira, given=Pedro Henrique, prefix=de useprefix=true family=Sousa Almeida, given=Joao Lucas, Rocco Sedona, Yanghui Kang, Srija Chakraborty, Sizhe Wang, Carlos Gomes, Ankur Kumar, Vishal Gaur, Myscon Truong, Denys Godwin, Sam Khallaghi, Hyunho Lee, Chia-Yu Hsu, Ata Akbari Asanjan, Besart Mujeci, Disha Shidham, Rufai Omowunmi Balogun, Venkatesh Kolluru, Trevor Keenan, Paulo Arevalo, Wenwen Li, Hamed Alemohammad, Pontus Olofsson, Timothy Mayer, Christopher Hain, Robert Kennedy, Bianca Zadrozny, David Bell, Gabriele Cavallaro, Campbell Watson, Manil Maskey, Rahul Ramachandran, and Juan Bernabe Moreno. 2026. Prithvi-EO-2.0: A Versatile Multitemporal Foundation Model for Earth Observation Applications. 64 (2026), 1–20. doi:10.1109/TGRS.2025.3642610

[22] Ningjing Wang, Xinyu Wang, Yang Pan, Wanqiang Yao, and Yanfei Zhong. 2025. WHU-RuR+: A Benchmark Dataset for Global High-Resolution Rural Road Extraction. 139 (2025), 104518. doi:10.1016/j.jag.2025.104518

[23] Xuan Wang, Xizhi Jin, Zhe Dai, Yuxuan Wu, and Abdellah Chehri. 2025. Deep Learning-Based Methods for Road Extraction From Remote Sensing Images: A Vision, Survey, and Future Directions. 13, 1 (2025), 55–78. doi:10.1109/MGRS.2024.3491014

[24] Michael A. Wulder, Thomas R. Loveland, David P. Roy, Christopher J. Crawford, Jeffrey G. Masek, Curtis E. Woodcock, Richard G. Allen, Martha C. Anderson, Alan S. Belward, Warren B. Cohen, John Dwyer, Angela Erb, Feng Gao, Patrick Griffiths, Dennis Helder, Txomin Hermosilla, James D. Hipple, Patrick Hostert, M. Joseph Hughes, Justin Huntington, David M. Johnson, Robert Kennedy, Ayse Kilic, Zhan Li, Leo Lymburner, Joel McCorkel, Nima Pahlevan, Theodore A. Scambos, Crystal Schaaf, John R. Schott, Yongwei Sheng, James Storey, Eric Vermote, James Vogelmann, Joanne C. White, Randolph H. Wynne, and Zhe Zhu. 2019. Current Status of Landsat Program, Science, and Applications. 225 (2019), 127–147. doi:10.1016/j.rse.2019.02.015

[25] Ibrahim Salihu Yusuf, Iffanice Houndayi, Rym Oualha, Mohamed Aziz Cherif, Kobby Panford-Quainoo, and Arnu Pretorius. 2025. *InstaGeo: Compute-Efficient Geospatial Machine Learning from Data to Deployment.* arXiv:2510.05617 [cs] doi:10.48550/arXiv.2510.05617

[26] Lichen Zhou, Chuang Zhang, and Ming Wu. 2018. D-LinkNet: LinkNet With Pretrained Encoder and Dilated Convolution for High Resolution Satellite Imagery Road Extraction. 182–186. https://openaccess.thecvf.com/content_cvpr_2018_workshops/w4/html/Zhou_D-LinkNet_LinkNet_With_CVPR_2018_paper.html